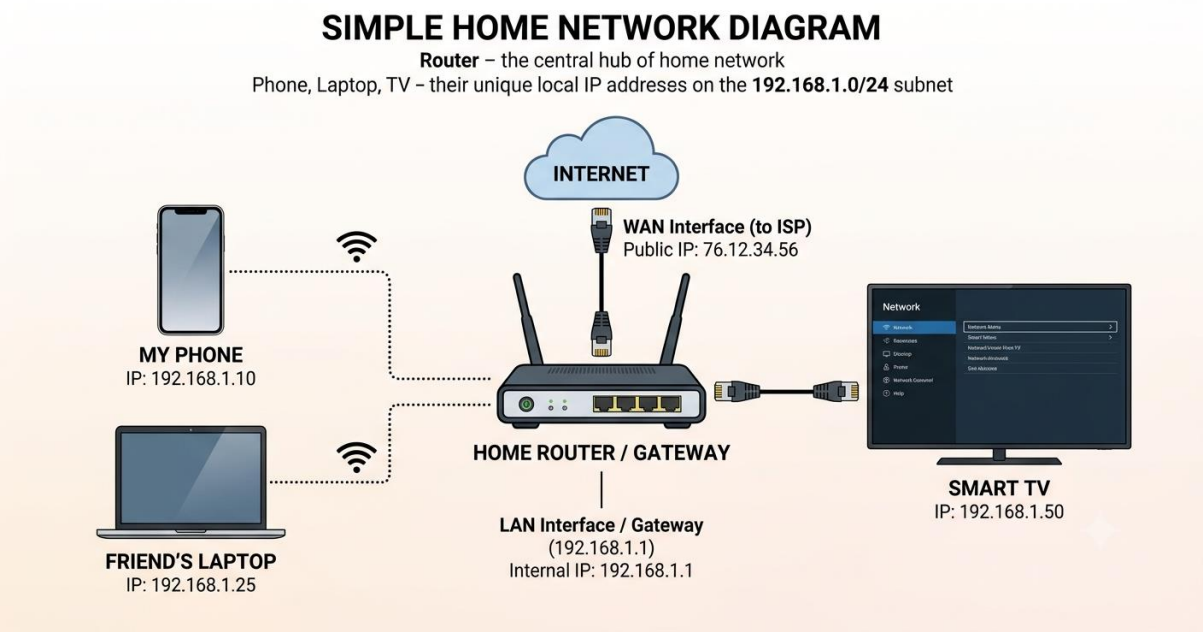
Ip Routing and Protocols

Task 1:- Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



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Task 2:- Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Network Setup & IP Addressing

- **Your Laptop (Source):** `192.168.10.15` (Subnet: `255.255.255.0`)
- **Your Home Router (Default Gateway / Interface 1):** `192.168.10.1`
- **Internet / Intermediate Routers:** Routing traffic across public hops (e.g., Gateway Router `203.0.113.1`)
- **Friend's Computer (Destination):** `172.16.20.45` (Subnet: `255.255.255.0`)

Step 1: Your Laptop Checks the Destination

When you initiate communication from your laptop (`192.168.10.15`) to your friend's computer (`172.16.20.45`), the laptop first performs a **subnet check** using its subnet mask (`255.255.255.0`).

- It compares its own network segment (`192.168.10.x`) with the destination (`172.16.20.45`).
- Because the destination is on a completely different network, the laptop realizes it cannot talk to the friend's computer directly via local Layer 2 broadcast. It must hand the packet over to its **Default Gateway** (your home router at `192.168.10.1`).

Routing Table Checked at Your Laptop:

Network Destination	Netmask	Gateway (Next Hop)	Interface
`192.168.10.0`	`255.255.255.0`	`0.0.0.0` (On-link)	`eth0` (Local)
`0.0.0.0` (Default)	`0.0.0.0`	`192.168.10.1`	`eth0`

- **Routing Decision:** The laptop matches the default route (`0.0.0.0`), encapsulates the packet into an Ethernet frame addressed to the router's MAC address, and sends it out to the home router.

Step 2: The Home Router Receives the Packet

Your home router receives the packet on its local LAN interface (`192.168.10.1`). It strips away the Layer 2 Ethernet header, inspects the network layer header, and looks at the destination IP (`172.16.20.45`).

Routing Table Checked at the Home Router:

Network Destination	Netmask	Gateway (Next Hop)	Interface
`192.168.10.0`	`255.255.255.0`	`0.0.0.0`	`LAN` (`192.168.10.1`)
`172.16.20.0`	`255.255.255.0`	`203.0.113.254` (ISP Gateway)	`WAN` (`76.12.34.56`)

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Network Destination	Netmask	Gateway (Next Hop)	Interface
`0.0.0.0` (Default)	`0.0.0.0`	`203.0.113.254`	`WAN`

- **Routing Decision:** The router matches the specific route for `172.16.20.0/24`. It rewrites the packet headers, changes the source/destination MAC addresses for wide-area transport, and forwards the packet out of its **WAN interface** toward the Internet Service Provider (ISP) gateway (`203.0.113.254`).

Step 3: Transit Across Intermediate Routers (The Internet Core)

As the packet traverses the internet, it hops through several carrier routers. Each intermediate router inspects the destination IP address (`172.16.20.45`), checks its routing table for the longest prefix match, and forwards the packet closer to its destination network.

Routing Table Checked at an Intermediate ISP Router:

Network Destination	Netmask	Gateway (Next Hop)	Interface
`172.16.0.0`	`255.255.0.0`	`198.51.100.14`	`Serial0/1`
`0.0.0.0` (Default)	`0.0.0.0`	`198.51.100.1`	`Serial0/0`

- **Routing Decision:** The router finds a match in the broader block `172.16.0.0/16` and pushes the packet to the next upstream router via its serial/fiber interface.

Step 4: Arrival at the Friend's Local Network Gateway

The packet finally reaches your friend's local network edge router, which handles the final destination network (`172.16.20.0/24`).

Routing Table Checked at the Friend's Edge Router:

Network Destination	Netmask	Gateway (Next Hop)	Interface
`172.16.20.0`	`255.255.255.0`	`0.0.0.0` (Directly Connected)	`LAN0` (`172.16.20.1`

Task 3:- Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

1. Direct Cable Connection (Crossover / Ethernet)

- **How it works:** Connect two devices directly using an Ethernet cable without any router in between.
- **Best for:** Transferring files quickly between two computers.
- **IP setup:** Both devices need manual static IP addresses in the same subnet (e.g., Device A: `192.168.1.1`, Device B: `192.168.1.2`) with the same subnet mask (`255.255.255.0`).

2. Wi-Fi Direct / AirDrop / Quick Share

- **How it works:** Devices create a temporary, direct point-to-point wireless connection using Wi-Fi and Bluetooth hardware without routing through an internet access point.
- **Best for:** Sending photos, documents, or files between a phone and a laptop instantly.
- **IP setup:** Handled automatically behind the scenes by the operating system.

3. Home Router (Local Area Network / LAN)

- **How it works:** Devices connect to the same Wi-Fi router or switch. The router acts as the central traffic controller, assigning local IP addresses using DHCP.
- **Best for:** Connecting multiple devices (phones, laptops, TVs) to share files, printers, or stream media locally.
- **IP setup:** Fully automated via DHCP (`192.168.x.x` or `10.x.x.x`), where the router serves as the default gateway.

Task 4:- Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing

Static routing involves manually configuring routes in a router's routing table. It requires an administrator to explicitly define every path that packets must take to reach different subnets.

- **Real-World Example:** A small standalone cloud kitchen with a single fixed broadband line connecting its Point-of-Sale (POS) terminal to a local kitchen display system (KDS). Because the network layout never changes, a static route ensures the billing terminal always talks directly to the kitchen printer through a fixed path.

Pros of Static Routing

- **Zero Overhead:** Routers use no CPU, memory, or bandwidth to calculate paths or exchange routing updates with neighbors, keeping network performance lean.
- **Higher Security:** Because no routing updates are broadcast across the network, there are no routing protocol packets for malicious actors to intercept or spoof.
- **Predictable Pathing:** Packets always follow the exact manual path specified by the administrator, making it easy to trace where traffic flows.

Cons of Static Routing

- **No Automatic Failover:** If a primary link or physical cable fails, traffic drops entirely unless an administrator manually logs in and switches to a backup route.
- **Poor Scalability:** Adding new subnets, branches, or dark stores requires manual updates to every single router in the network, which becomes unmanageable as an organization grows.
- **High Administrative Burden:** Network changes demand constant manual oversight, increasing the risk of human configuration errors (like typos in IP addresses).

Dynamic Routing (e.g., OSPF, RIP, BGP)

Dynamic routing uses protocols where routers automatically discover network topology, share routing information with neighboring routers, and recalculate the best paths on the fly.

- **Real-World Example:** A nationwide food delivery app (like Zomato or Swiggy) managing thousands of orders per second across multiple regional cloud hubs, data centres, and dark stores. If a primary fiber-optic link between a major cloud region and a regional hub goes down, OSPF or BGP protocols instantly reroute order data through an alternate path without human intervention.

Pros of Dynamic Routing

- **Automatic Fault Tolerance:** If a link drops or a router fails, the protocol automatically recalculates an available alternate path within seconds, ensuring high availability.

- **High Scalability:** When a company opens a new branch or adds a subnet, the routing table updates itself automatically without requiring manual configuration on existing routers.
- **Efficient Traffic Management:** Advanced protocols can dynamically select paths based on real-time metrics like link cost, bandwidth, or latency to optimize performance.

Cons of Dynamic Routing

- **Resource Consumption:** Routers must dedicate CPU, memory, and background network bandwidth to run routing daemons and exchange frequent "hello" and update packets.
- **Security Vulnerabilities:** If routing protocols are not properly authenticated and secured, attackers can inject false routing updates (route poisoning or hijacking), redirecting sensitive traffic.
- **Complex Troubleshooting:** Because paths can shift dynamically based on network conditions, diagnosing intermittent latency or packet loss can be significantly harder for network engineers.